

Andrew Bai

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EDUCATION

University of Waterloo

Waterloo, Ontario

Bachelor of Applied Science in Mechatronics Engineering, GPA: 85%

September 2025 - June 2030

- **Relevant Coursework:** Dynamics, Circuit Analysis, Data Structures & Algorithms, Numerical Methods, Material Science

TECHNICAL SKILLS

Programming & Software: ROS2, SLAM, Python, C++, JavaScript, React, pandas, NumPy, Matplotlib, OpenCV

Embedded Systems & Hardware: ESP32, Arduino, Circuit Design, Soldering, SolidWorks, 3D printing

Developer Tools: Git, GitLab, Docker, VS Code, Jira, CI/CD pipelines, Agile/Scrum, component-level testing, LaTeX

EXPERIENCE

Onboard Payload Software Co-op

May 2026 - August 2026

Telesat

Ottawa, Ontario

- Shipped production code in C++ and wrote component level tests in Python for Telesat's low earth orbit satellite constellation.
- Supported development of message intake logic, MODCOD control loop, and debug command line interface, as well as many other general maintenance/debugging tasks.
- Collaborated in an Agile/Scrum environment, managed tasks and merge requests in Jira and GitLab, including CI/CD pipeline usage.

General Autonomy Member

January 2026 - Present

Waterloo Aerial Robotics Group

Waterloo, Ontario

- Participated as a general autonomy member for the group's competition drone.
- Implemented HSV-based color detection in Python and Open CV
- Worked on a multi-process drone ground station in Python using MAVLink messaging and inter process queues to communicate with and control a simulated drone.

PROJECTS

3D Room Scanner

August 2026 - September 2026

- Designed and built a 3D room scanner that fuses a spinning Unitree L2 LiDAR with an IMU through a Point-LIO SLAM pipeline in ROS2
- Modelled both sensors in Gazebo using their datasheet specs and injected realistic noise and IMU bias drift to validate SLAM accuracy before touching hardware
- 3D printed the sensor chassis and field-tested the scanner, maintaining accurate maps at movement speeds up to 5 km/h and rotations of up to 90°/s.

Hexapod Robot

April 2026 - July 2026

- Designed and 3D printed a fully custom six-legged hexapod robot from scratch in SolidWorks, modeling all mechanical components and leg linkages
- Programmed leg kinematics and gait control in C++ on an ESP32 microcontroller, achieving stable coordinated walking at speeds up to 0.1 m/s
- Designed circuit diagrams and wired the full electrical system by hand, including soldering all connections between the ESP32, servos, and power distribution

Automatic Poker Dealer

October 2025 - November 2025

- Collaborated with three teammates to design and build a poker dealing robot.
- Programmed game logic, sensor inputs, and card dispensing in C++ using the VEX IDE.
- Integrated a color sensor to intake bets, buttons to progress the game, and a distance sensor to ensure consistent card dealing.